

28.09.2026

PRACTICAL - 10(a)

INTERNETWORKING WITH ROUTERS

AIM:-

Internetworking with routers in CISCO PACKET TRACER simulator.

PROCEDURE:-

STEP 1:- Configuring Router 1

- 1.) Open cisco packet tracer and place one router and 2 PCs.
- 2.) Select the router and open the CLI. and press enter to access the command prompt.
- 3.) Enter privileged mode using enable.
- 4.) Enter global configuration mode using configure terminal.
- 5.) configure FastEthernet0/0 with:
 - IP address : 192.168.10.1
 - Subnet mask : 255.255.255.0
- 6.) Use no shutdown to activate the interface.
- 7.) configure FastEthernet0/1 with:
 - IP address : 192.168.20.1
 - Subnet mask : 255.255.255.0
- 8.) Use no shutdown to activate the interface.

STEP 2:- Configuring PC's.

Device	IP Address	Subnet mask	Default gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

STEP 3:- Connecting PC's with Router.

- 1.) Connect PC0 FastEthernet0 to Router1 FastEthernet0/0 using a copper straight-through cable.
- 2.) Connect PC1 FastEthernet0 to Router1 FastEthernet0/1 using a copper straight-through cable.
- 3.) Wait until the interfaces become active.
- 4.) Use simulation mode and send a simple PDU from PC0 to PC1.
- 5.) Verify that the PDU reaches PC1 and an acknowledgment is received back to PC0.

CONFIGURATION TABLE:-

Router Configuration

Device	FastEthernet 0/0	Subnet Mask	FastEthernet 0/1	Subnet Mask
Router1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

```

PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
    
```

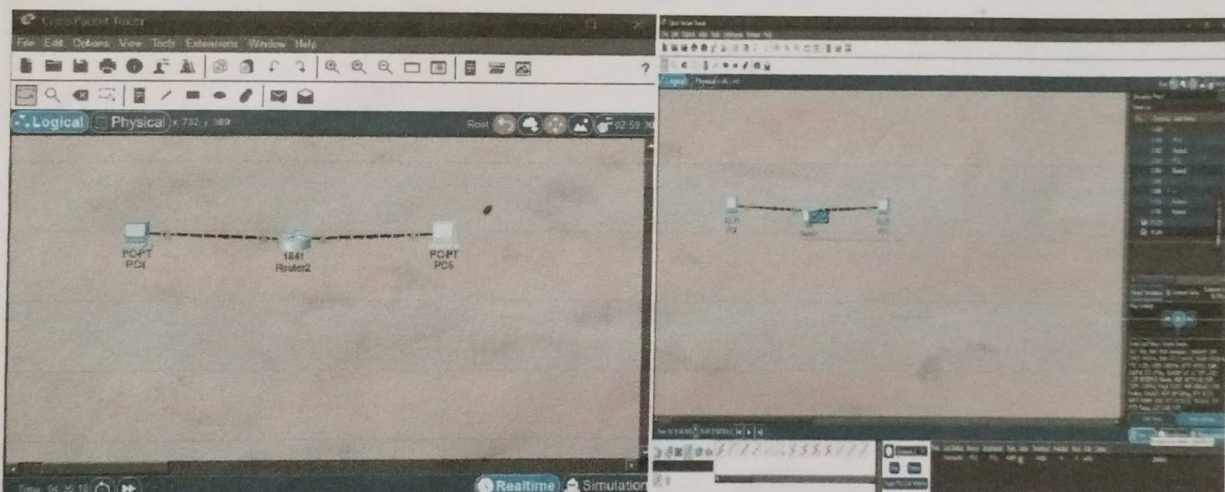
```

PC5
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
    
```



Scenario 0	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
		Successful	PC4	PC5	ICMP		0.000	N	0	(edit)	(delete)

RESULT:-

Thus, a simple internetwork was successfully designed and configured using a router in Cisco Packet Tracer.

28.09.2026

PRACTICAL - 10(b) INTERNETWORK USING WIRELESS ROUTER

AIM:-

To design and configure an internetwork using a wireless router, HTTP server and Internet cloud in Cisco Packet Tracer.

PROCEDURE:-

STEP 1:- Build the network topology.

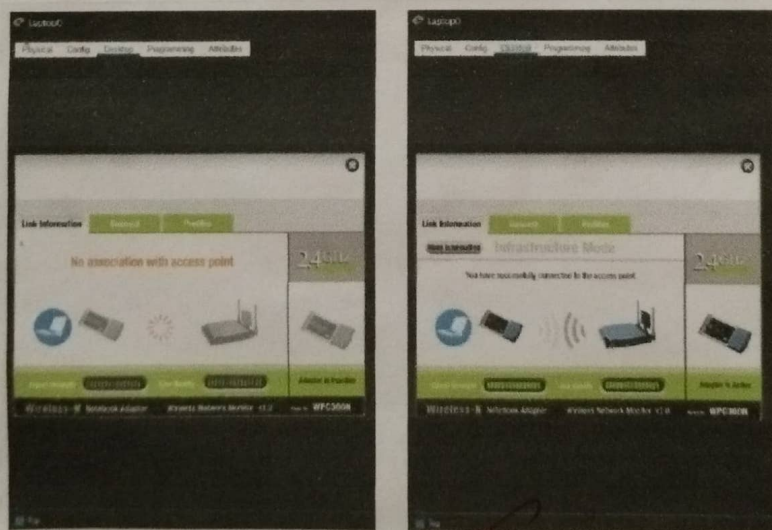
- 1) Open Cisco Packet Tracer and add the devices like PC, Router, Laptop, cable modem, Internet cloud and Server.
- 2) Connect the devices using the appropriate cables.

STEP 2:- Configure the Router.

- 1) Wireless Router → GUI → Wireless.
Network name → HomeNetwork
- 2) Go to Setup tab and ensure DHCP is enabled.
- 3) Configure the server as 208.67.220.220.
- 4) Save the settings.

STEP 3:- Configure the laptop.

- 1) Laptop → Physical tab → poweroff → Remove ethernet cable and add WPC300N wireless module → Power on the laptop
- 2) Desktop → PC wireless. select connect tab and select HomeNetwork.
- 3) Connect the laptop to the wireless network.



As designed, all devices are successfully connected to the wireless network and the server is accessible from the laptop.

STEP 4:- Configure the PC.

- 1) PC → Desktop → IP configuration. Select DHCP.
- 2) The PC automatically receives an IP address from the wireless router.
- 3) Use the command prompt and ipconfig /all to verify the configuration.

STEP 5:- Configure the Internet cloud :

- 1) open the Internet cloud
- 2) Config → cable
- 3) set :- From port :- Coaxial To port : Ethernet
- 4) click Add.
- 5) Go to Ethernet and select cable as Provider network.

STEP 6:- Configure the Cisco.com server

DHCP configuration:-

Parameter	Value
Pool Name	DHCP pool
Default gateway	208.67.220.220
DNS server	208.67.220.220
Starting IP	208.67.220.1
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add.

DNS Configuration:-

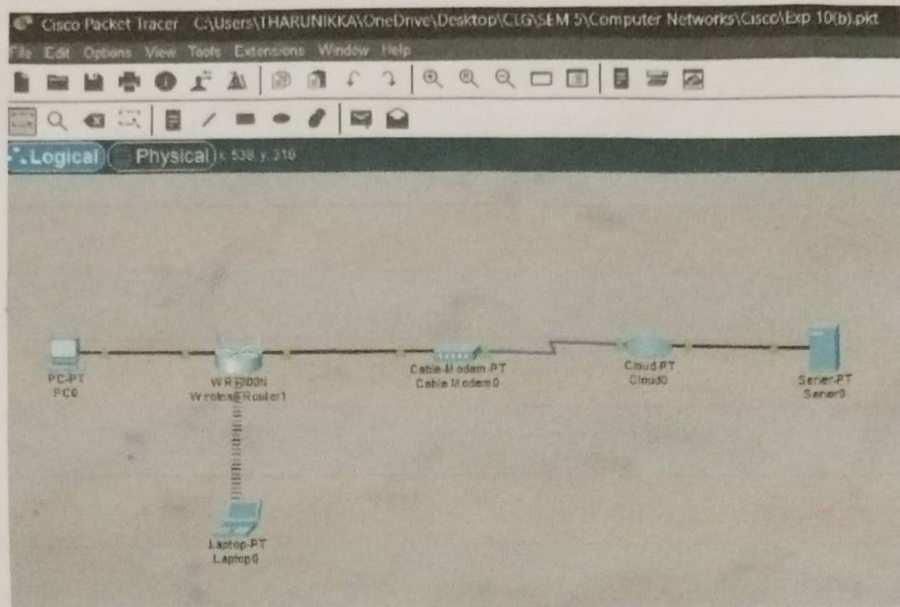
Parameter	Value
Name	cisco.com
Type	A Record
Address	208.67.220.220

Click Add.

STEP 7:-

Verify connectivity

Lab 10: DHCP and DNS configuration on Cisco routers and switches. The lab is completed successfully. All devices are connected and configured correctly. The DHCP server is running and the DNS server is also running. The lab is completed successfully.



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 208.67.220.220

Pinging 208.67.220.220 with 32 bytes of data:

Request timed out.
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=9ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 3ms
```

RESULT:-

Thus the wireless router, DHCP server and Internet cloud were successfully configured in Cisco Packet Tracer and connectivity was verified successfully.